

# MOHAMMAD TALHA

Dhaka, Bangladesh | +8801835064233 | talhajubayer78@gmail.com | [linkedin.com/in/mohammad-talha](https://www.linkedin.com/in/mohammad-talha) | [Portfolio](#)

I am a CSE student with a passion for problem-solving and technology, eager to tackle dynamic challenges and apply analytical skills to real-world applications. I have demonstrated proficiency in data analysis, programming, and event management, along with strong collaboration and leadership abilities.

## Area of Expertise

|                               |                                 |                    |
|-------------------------------|---------------------------------|--------------------|
| Data Analysis & Visualization | Event Management & Coordination | Web Development    |
| Team Leadership               | Communication & Documentation   | Programming        |
| Database Management           | Microsoft Office Suite          | Shift Coordination |

## Achievement

### Champion on ULAB EEE Fest-2026

- Won the ULAB EEE Fest-2026 for an IoT-based Smart Safety Helmet.

### ULAB CSE Project Competition- 2026

- Our project Notun Alo successfully took the 4th position among 55 projects in the ULAB CSE project competition.

### Certified Data Analyst

- Completed Data Analysis certification from DataSolution360 Institute.

## Projects

### Notun Alo - Smart AI Waste Management & Recycling Platform

- Built a dual-engine hybrid web platform connecting households with recycling agencies through an automated **AI-powered agency** assignment system using a multi-factor weighted scoring algorithm.
- Engineered a Gradient Boosting ML model (Scikit-learn) for pickup ETA prediction, combined with K-Means unsupervised clustering to divide Dhaka into optimized operational service zones.
- Designed a 3-tier fault-tolerant failover architecture ensuring zero user-facing downtime across full ML operation, degraded mode, and native PHP fallback states.
- Integrated a Bengali-first conversational AI **chatbot** (Claude API) with real-time user history access, enabling natural language pickup scheduling with automatic database insertion on intent detection.
- Implemented a **gamified reward system** with points-based redemption, badges, and leaderboard to drive sustained recycling engagement across urban households
- *Tech Stack: PHP, Python, Flask, MySQL, Scikit-learn, JavaScript, HTML5, CSS, Claude API*

### IoT-Based Smart Safety Helmet

- Developed an IoT-enabled smart helmet using ESP32 with alcohol, drowsiness, crash, and helmet-wearing detection for active rider safety.
- Integrated automated engine cutoff, buzzer alerts, and real-time safety monitoring using Blynk and Firebase.
- *Tech Stack: ESP32, Arduino, IoT, MPU6050, MQ-3, MAX30102, Blynk, Firebase*

### Fish Recommendation System (AquaRec)

- Developed an AI-powered fish recommendation system using Random Forest & KNN with SMOTE, suggesting species based on water quality (pH, temperature, turbidity).
- Built a Streamlit web app for real-time predictions, improving aquaculture decision-making and yield.
- *Tech Stack: Python, Streamlit, Scikit-learn, Joblib, MySQL, HTML5, CSS*

### Exam Management System

- Built a dual-portal (Admin/Student) system with secure question management, result tracking, login, profile updates, and instant exam results.
- *Tech Stack: Java, MySQL*

## Volunteer Experience

### Campus Ambassador – Kaizen SUST (OmniStart 1.0)

Oct 2025 – Jan 2026

- Promoted events, increasing student engagement and registrations through campus outreach.
- Acted as liaison, managing communication between participants and organizing teams.

### Event Admin – MIND SHIFT-2025

Jul 2025 – Aug 2025

- Coordinated event operations and handled administrative tasks for smooth execution.
- Managed participant data and ensured effective team communication.

### Campus Ambassador – ICT Olympiad Bangladesh S2

Oct 2024 – Feb 2025

- Led campus campaigns to boost awareness and participation.
- Supported event execution and handled participant coordination.

#### **Event Admin – Idea Hunters 4.0**

*Sept 2024 – Jan 2025*

- Managed event planning and operations for large-scale activities.
- Oversaw team coordination and ensured smooth event scheduling.

#### **Volunteer – ULAB Communication Office**

*Jan 2024 – Dec 2024*

- Assisted in event organization, content creation, and documentation.
- Supported internal communication and coordination for university programs.

#### **Organizing Secretary – MISHAS (College Organization)**

*Jan 2018 – Jan 2019*

- Led event planning and execution for college activities.
- Managed team coordination and communication.

### **Internship**

---

#### **ICT Olympiad Bangladesh S3 (Course Curriculum Team)**

*Sept 2025 – Jan 2026*

- Managed class hosting, recordings, and video editing for course delivery.
- Organized Q&A datasets, assisted in curriculum development, and coordinated with instructors.

### **Education**

---

#### **University of Liberal Arts Bangladesh**

*Sept 2022 – Present*

*Bachelor in CSE*

- Current CGPA more than 3.5.
- Successfully completed projects on Java, Python, MySQL, and Frontend.

#### **Tamirul Millat Kamil Madrasah**

*2019 – 2021*

*Arabic Medium*

- Finished Higher School Certificate in 2021 with 5.00 GPA.
- Champion (twice) in cricket tournament.

#### **Tamirul Millat Kamil Madrasah**

*2017 – 2019*

*Arabic Medium*

- Finished Senior School Certificate in 2019 with 5.00 GPA.

### **Additional Information**

---

- **Languages:** Bengali (Native), English (Professional Fluency), Hindi.
- **Technical Skills:** Microsoft Suite (Word, Excel, PowerPoint, Power BI), Programming (HTML5, CSS, Java, SQL, Python), Video Editing (CapCut).